

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Camera
Manufacturer: AVer
Firmware Version: N/A
Model(s): TR311HN, TR313, TR333, TR310

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.15.0000-b004	2.14.0

Version History

Module Version	Date	Notes
1_2_0_0	3/3/2022	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- For Serial control, DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '7'.
Example: `InterfaceName.DeviceID = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='TR311HN')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='TR311HN')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 52381, Model='TR311HN')</code>

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutoExposure	Value 'Full Auto' 'Iris Priority'	Value 'Manual' 'Bright'	Value 'Shutter Priority'
# AutoExposure example InterfaceName.Set('AutoExposure', 'Full Auto')			
Command AutoTracking	Value 'On'	Value 'Off'	
# AutoTracking example InterfaceName.Set('AutoTracking', 'On')			
Command Backlight	Value 'On'	Value 'Off'	
# Backlight example InterfaceName.Set('Backlight', 'On')			
Command FocusMode	Value 'Auto'	Value 'Manual'	
# FocusMode example InterfaceName.Set('FocusMode', 'Auto')			
Command Gain	Value 'Up'	Value 'Down'	
# Gain example InterfaceName.Set('Gain', 'Up')			
Command Home	Value None		
# Home example InterfaceName.Set('Home', None)			
Command Iris	Value 'Up'	Value 'Down'	
# Iris example InterfaceName.Set('Iris', 'Up')			
Command PanTilt	Value 'Up' 'Right' 'Down Left'	Value 'Down' 'Up Left' 'Down Right'	Value 'Left' 'Up Right' 'Stop'
Qualifier Key 'Pan Speed'	Qualifier Value '1' – '24'		
Qualifier Key 'Tilt Speed'	Qualifier Value '1' – '24'		
# PanTilt example InterfaceName.Set('PanTilt', 'Up', {'Pan Speed': '1', 'Tilt Speed': '1'})			
Command PanTiltReset	Value None		
# PanTiltReset example InterfaceName.Set('PanTiltReset', None)			

Global Scripter Module Communication Sheet

Command PeopleSize	Value 'Full Body'	Value 'Upper Body'	
# PeopleSize example InterfaceName.Set('PeopleSize', 'Full Body')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command PresetRecall	Value '0' – '94'	Value '96' – '159'	Value '167' – '255'
# PresetRecall example InterfaceName.Set('PresetRecall', '0')			
Command PresetReset	Value '0' – '94'	Value '96' – '159'	Value '167' – '255'
# PresetReset example InterfaceName.Set('PresetReset', '0')			
Command PresetSave	Value '0' – '94'	Value '96' – '159'	Value '167' – '255'
# PresetSave example InterfaceName.Set('PresetSave', '0')			
Command Shutter	Value 'Up'	Value 'Down'	
# Shutter example InterfaceName.Set('Shutter', 'Up')			
Command Switch	Value None		
# Switch example InterfaceName.Set('Switch', None)			
Command TrackingMode	Value 'Presenter' ¹	Value 'Zone' ¹	Value 'Hybrid' ²
# TrackingMode example InterfaceName.Set('TrackingMode', 'Presenter')			
Command TrackingPoint	Value None		
# TrackingPoint example InterfaceName.Set('TrackingPoint', None)			
Command Zoom	Value 'Tele'	Value 'Wide'	Value 'Stop'
Qualifier Key 'Zoom Speed'	Qualifier Value '0' – '7'		
# Zoom example InterfaceName.Set('Zoom', 'Tele', {'Zoom Speed': '0'})			

¹ For firmware 0.25 or above.

² For firmware 0.35 or above.

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AutoExposure	'Full Auto' 'Iris Priority'	'Manual' 'Bright'	'Shutter Priority'
# AutoExposure example InterfaceName.Update('AutoExposure') Value = InterfaceName.ReadStatus('AutoExposure') InterfaceName.SubscribeStatus('AutoExposure', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	
FocusMode	'Auto'	'Manual'	
# FocusMode example InterfaceName.Update('FocusMode') Value = InterfaceName.ReadStatus('FocusMode') InterfaceName.SubscribeStatus('FocusMode', None, FeedbackHandler)			
Command	Value	Value	
Power	'On'	'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutoExposure	Value 'Full Auto' 'Iris Priority'	Value 'Manual' 'Bright'	Value 'Shutter Priority'
# AutoExposure example InterfaceName.Set('AutoExposure', 'Full Auto')			
Command AutoTracking	Value 'On'	Value 'Off'	
# AutoTracking example InterfaceName.Set('AutoTracking', 'On')			
Command Backlight	Value 'On'	Value 'Off'	
# Backlight example InterfaceName.Set('Backlight', 'On')			
Command FocusMode	Value 'Auto'	Value 'Manual'	
# FocusMode example InterfaceName.Set('FocusMode', 'Auto')			
Command Gain	Value 'Up'	Value 'Down'	
# Gain example InterfaceName.Set('Gain', 'Up')			
Command Home	Value None		
# Home example InterfaceName.Set('Home', None)			
Command Iris	Value 'Up'	Value 'Down'	
# Iris example InterfaceName.Set('Iris', 'Up')			
Command PanTilt	Value 'Up' 'Right' 'Down Left'	Value 'Down' 'Up Left' 'Down Right'	Value 'Left' 'Up Right' 'Stop'
Qualifier Key 'Pan Speed'	Qualifier Value '1' – '24'		
Qualifier Key 'Tilt Speed'	Qualifier Value '1' – '24'		
# PanTilt example InterfaceName.Set('PanTilt', 'Up', {'Pan Speed': '1', 'Tilt Speed': '1'})			
Command PanTiltReset	Value None		
# PanTiltReset example InterfaceName.Set('PanTiltReset', None)			

Command PeopleSize	Value 'Full Body'	Value 'Upper Body'	
# PeopleSize example InterfaceName.Set('PeopleSize', 'Full Body')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command PresetRecall	Value '0' – '94'	Value '96' – '159'	Value '167' – '255'
# PresetRecall example InterfaceName.Set('PresetRecall', '0')			
Command PresetReset	Value '0' – '94'	Value '96' – '159'	Value '167' – '255'
# PresetReset example InterfaceName.Set('PresetReset', '0')			
Command PresetSave	Value '0' – '94'	Value '96' – '159'	Value '167' – '255'
# PresetSave example InterfaceName.Set('PresetSave', '0')			
Command Shutter	Value 'Up'	Value 'Down'	
# Shutter example InterfaceName.Set('Shutter', 'Up')			
Command Switch	Value None		
# Switch example InterfaceName.Set('Switch', None)			
Command TrackingMode	Value 'Presenter' ¹	Value 'Zone' ¹	Value 'Hybrid' ²
# TrackingMode example InterfaceName.Set('TrackingMode', 'Presenter')			
Command TrackingPoint	Value None		
# TrackingPoint example InterfaceName.Set('TrackingPoint', None)			
Command Zoom	Value 'Tele'	Value 'Wide'	Value 'Stop'
Qualifier Key 'Zoom Speed'	Qualifier Value '0' – '7'		
# Zoom example InterfaceName.Set('Zoom', 'Tele', {'Zoom Speed': '0'})			

¹ For firmware 0.25 or above.

² For firmware 0.35 or above.

Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AutoExposure	Value 'Full Auto' 'Iris Priority'	Value 'Manual' 'Bright'	Value 'Shutter Priority'
# AutoExposure example InterfaceName.Update('AutoExposure') Value = InterfaceName.ReadStatus('AutoExposure') InterfaceName.SubscribeStatus('AutoExposure', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command FocusMode	Value 'Auto'	Value 'Manual'	
# FocusMode example InterfaceName.Update('FocusMode') Value = InterfaceName.ReadStatus('FocusMode') InterfaceName.SubscribeStatus('FocusMode', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Male Mini DIN9 RS-232 Serial Cable

Notes for the Device

To ensure the functionality of the driver, please set the following:

- Menu → Advanced Setting → Control → Type → RS-232
- Menu → Advanced Setting → Control → Protocol → VISCA

Serial communication

Port Type: RS-232

Baud Rate: 9600

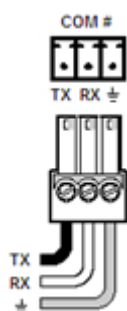
Data Bits: 8

Parity: None

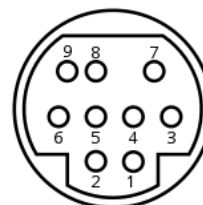
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Please contact the manufacturer for pin assignment diagram.



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface.

Port Type: Ethernet (UDP)

Device Listen Port: 52381

Device Reply Port: 52381

Logon Credentials Supported: No

Multi-Connection Capabilities: Undetermined

Listen Port Changeability: No

Listen Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

When using multiple UDP modules within the same configuration each driver should have a unique 'Reply on Port' number due to the nature of UDP communication. If you have a module with a fixed 'Reply on Port' number, only one instance of the module can be used within the configuration.

Appendix A. Set Commands(Serial)

Auto Exposure Bright	\x81\x01\x049\x0D\xff
Auto Exposure Full Auto	\x81\x01\x049\x00\xff
Auto Exposure Iris Priority	\x81\x01\x049\x0B\xff
Auto Exposure Manual	\x81\x01\x049\x03\xff
Auto Exposure Shutter Priority	\x81\x01\x049\x0A\xff
Auto Tracking Off	\x81\x01\x04}\x03\xff
Auto Tracking On	\x81\x01\x04}\x02\xff
Backlight Off	\x81\x01\x043\x03\xff
Backlight On	\x81\x01\x043\x02\xff
Focus Mode Auto	\x81\x01\x048\x02\xff
Focus Mode Manual	\x81\x01\x048\x03\xff
Gain Down	\x81\x01\x04\x0C\x03\xff
Gain Up	\x81\x01\x04\x0C\x02\xff
Home None	\x81\x01\x06\x04\xff
Iris Down	\x81\x01\x04\x0B\x03\xff
Iris Up	\x81\x01\x04\x0B\x02\xff
Pan Tilt Down Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x01\x02\xff
Pan Tilt Down Left Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x01\x02\xff
Pan Tilt Down Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x02\xff
Pan Tilt Down Left Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x01\x02\xff
Pan Tilt Down Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x02\xff
Pan Tilt Down Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x03\x02\xff
Pan Tilt Down Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x02\xff
Pan Tilt Down Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x03\x02\xff
Pan Tilt Down Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x02\xff
Pan Tilt Down Right Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x02\x02\xff
Pan Tilt Down Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x02\xff
Pan Tilt Down Right Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x02\x02\xff
Pan Tilt Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x01\x03\xff
Pan Tilt Left Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x01\x03\xff
Pan Tilt Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x03\xff
Pan Tilt Left Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x01\x03\xff
Pan Tilt Reset None	\x81\x01\x06\x05\xff
Pan Tilt Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x03\xff
Pan Tilt Right Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x02\x03\xff
Pan Tilt Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x03\xff
Pan Tilt Right Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x02\x03\xff
Pan Tilt Stop Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x03\xff
Pan Tilt Stop Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x03\x03\xff
Pan Tilt Stop Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x03\xff
Pan Tilt Stop Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x03\x03\xff
Pan Tilt Up Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x01\x01\xff

Global Scriptor Module Communication Sheet

Revision: 3/3/2022

Pan Tilt Up Left Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x01\x01\xff
Pan Tilt Up Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x01\xff
Pan Tilt Up Left Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x01\x01\xff
Pan Tilt Up Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x01\xff
Pan Tilt Up Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x03\x01\xff
Pan Tilt Up Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x01\xff
Pan Tilt Up Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x03\x01\xff
Pan Tilt Up Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x01\xff
Pan Tilt Up Right Pan Speed 1 Tilt Speed 24	\x81\x01\x06\x01\x01\x18\x02\x01\xff
Pan Tilt Up Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x01\xff
Pan Tilt Up Right Pan Speed 24 Tilt Speed 24	\x81\x01\x06\x01\x18\x18\x02\x01\xff
People Size Full Body	\x81\x01\x04?\x01\xA0\xff
People Size Upper Body	\x81\x01\x04?\x01\xA1\xff
Power Off	\x81\x01\x04\x00\x03\xff
Power On	\x81\x01\x04\x00\x02\xff
Preset Recall 0	\x81\x01\x04?\x02\x00\xff
Preset Recall 159	\x81\x01\x04?\x02\x9F\xff
Preset Recall 167	\x81\x01\x04?\x02\xA7\xff
Preset Recall 255	\x81\x01\x04?\x02\xff\xff
Preset Recall 94	\x81\x01\x04?\x02^\xff
Preset Recall 96	\x81\x01\x04?\x02` \xff
Preset Reset 0	\x81\x01\x04?\x00\x00\xff
Preset Reset 159	\x81\x01\x04?\x00\x9F\xff
Preset Reset 167	\x81\x01\x04?\x00\xA7\xff
Preset Reset 255	\x81\x01\x04?\x00\xff\xff
Preset Reset 94	\x81\x01\x04?\x00^\xff
Preset Reset 96	\x81\x01\x04?\x00` \xff
Preset Save 0	\x81\x01\x04?\x01\x00\xff
Preset Save 159	\x81\x01\x04?\x01\x9F\xff
Preset Save 167	\x81\x01\x04?\x01\xA7\xff
Preset Save 255	\x81\x01\x04?\x01\xff\xff
Preset Save 94	\x81\x01\x04?\x01^\xff
Preset Save 96	\x81\x01\x04?\x01` \xff
Shutter Down	\x81\x01\x04\x0A\x03\xff
Shutter Up	\x81\x01\x04\x0A\x02\xff
Switch None	\x81\x01\x04?\x01\xA3\xff
Tracking Mode Hybrid	\x81\x01\x04?\x01\xA6\xff
Tracking Mode Presenter	\x81\x01\x04?\x01\xA4\xff
Tracking Mode Zone	\x81\x01\x04?\x01\xA5\xff
Tracking Point None	\x81\x01\x04?\x01\xA2\xff
Zoom Stop Zoom Speed 0	\x81\x01\x04\x07\x00\xff
Zoom Stop Zoom Speed 7	\x81\x01\x04\x07\x00\xff
Zoom Tele Zoom Speed 0	\x81\x01\x04\x07 \xff
Zoom Tele Zoom Speed 7	\x81\x01\x04\x07' \xff

Global Scriptor Module Communication Sheet

Zoom Wide Zoom Speed 0	\x81\x01\x04\x070\xff
Zoom Wide Zoom Speed 7	\x81\x01\x04\x077\xff

Appendix B. Update Commands(Serial)

Auto Exposure	\x81\x09\x049\xff
Focus Mode	\x81\x09\x048\xff
Power	\x81\x09\x04\x00\xff

Appendix C. Set Commands(Ethernet)

Auto Exposure Full Auto	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x049\x00\xFF
Auto Exposure Iris Priority	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x049\x0B\xFF
Auto Exposure Manual	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x049\x03\xFF
Auto Exposure Shutter Priority	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x049\x0A\xFF
Auto Tracking Off	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04}\x03\xFF
Auto Tracking On	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04}\x02\xFF
Backlight Off	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x043\x03\xFF
Backlight On	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x043\x02\xFF
Focus Mode Auto	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x048\x02\xFF
Focus Mode Manual	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x048\x03\xFF
Gain Down	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0C\x03\xFF
Gain Up	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0C\x02\xFF
Home None	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x04\xFF
Iris Down	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0B\x03\xFF
Iris Up	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0B\x02\xFF
Pan Tilt Down Left Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x02\xFF
Pan Tilt Down Left Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x01\x02\xFF
Pan Tilt Down Left Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x01\x02\xFF
Pan Tilt Down Left Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x01\x02\xFF
Pan Tilt Down Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x03\x02\xFF
Pan Tilt Down Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x03\x02\xFF
Pan Tilt Down Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x03\x02\xFF
Pan Tilt Down Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x03\x02\xFF
Pan Tilt Down Right Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x02\x02\xFF
Pan Tilt Down Right Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x02\x02\xFF
Pan Tilt Down Right Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x02\x02\xFF
Pan Tilt Down Right Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x02\x02\xFF
Pan Tilt Left Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x03\xFF
Pan Tilt Left Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x01\x03\xFF
Pan Tilt Left Pan Speed 24 Tilt	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x01\x03\xFF

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Speed 1	
Pan Tilt Left Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x01\x03\xFF
Pan Tilt Reset None	\x01\x00\x00\x05\x00\x00\x00\x01\x81\x01\x06\x05\xFF
Pan Tilt Right Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x02\x03\xFF
Pan Tilt Right Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x02\x03\xFF
Pan Tilt Right Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x02\x03\xFF
Pan Tilt Right Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x02\x03\xFF
Pan Tilt Stop Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x03\x03\xFF
Pan Tilt Stop Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x03\x03\xFF
Pan Tilt Stop Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x03\x03\xFF
Pan Tilt Stop Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x03\x03\xFF
Pan Tilt Up Left Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x01\x01\xFF
Pan Tilt Up Left Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x01\x01\xFF
Pan Tilt Up Left Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x01\x01\xFF
Pan Tilt Up Left Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x01\x01\xFF
Pan Tilt Up Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x03\x01\xFF
Pan Tilt Up Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x03\x01\xFF
Pan Tilt Up Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x03\x01\xFF
Pan Tilt Up Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x03\x01\xFF
Pan Tilt Up Right Pan Speed 1 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x01\x02\x01\xFF
Pan Tilt Up Right Pan Speed 1 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x01\x18\x02\x01\xFF
Pan Tilt Up Right Pan Speed 24 Tilt Speed 1	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x01\x02\x01\xFF
Pan Tilt Up Right Pan Speed 24 Tilt Speed 24	\x01\x00\x00\x09\x00\x00\x00\x01\x81\x01\x06\x01\x18\x18\x02\x01\xFF
People Size Full Body	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01\xA0\xFF
People Size Upper Body	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01\xA1\xFF
Power Off	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x00\x03\xFF

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Power On	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x00\x02\xff
Preset Recall 0	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x02\x00\xff
Preset Recall 159	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x02\x9F\xff
Preset Recall 167	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x02\xA7\xff
Preset Recall 255	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x02\xff\xff
Preset Recall 94	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x02^\xff
Preset Recall 96	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x02` \xff
Preset Reset 0	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x00\x00\xff
Preset Reset 159	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x00\x9F\xff
Preset Reset 167	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x00\xA7\xff
Preset Reset 255	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x00\xff\xff
Preset Reset 94	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x00^\xff
Preset Reset 96	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x00` \xff
Preset Save 0	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01\x00\xff
Preset Save 159	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01\x9F\xff
Preset Save 167	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01\xA7\xff
Preset Save 255	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01\xff\xff
Preset Save 94	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01^\xff
Preset Save 96	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01` \xff
Shutter Down	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0A\x03\xff
Shutter Up	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x0A\x02\xff
Switch None	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01xA3\xff
Tracking Mode Hybrid	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01xA6\xff
Tracking Mode Presenter	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01xA4\xff
Tracking Mode Zone	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01xA5\xff
Tracking Point None	\x01\x00\x00\x07\x00\x00\x00\x01\x81\x01\x04?\x01xA2\xff
Zoom Stop Zoom Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x07\x00\xff
Zoom Stop Zoom Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x07\x00\xff
Zoom Tele Zoom Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x07 \xff
Zoom Tele Zoom Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x07' \xff
Zoom Wide Zoom Speed 0	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x070\xff
Zoom Wide Zoom Speed 7	\x01\x00\x00\x06\x00\x00\x00\x01\x81\x01\x04\x077\xff

Appendix D. Update Commands(Ethernet)

Auto Exposure	\x01\x10\x00\x05\x00\x00\x00\x01\x81\x09\x049\xFF
Focus Mode	\x01\x10\x00\x05\x00\x00\x00\x01\x81\x09\x048\xFF
Power	\x01\x10\x00\x05\x00\x00\x00\x01\x81\x09\x04\x00\xFF